

RAMSES Mission
Scientific Gap Assessment, Payloads Study,
and Payload Enhancement Recommendation
for Asteroid (99942) Apophis

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Executive Summary

Following the selection of RAMSES as the reference mission, this study conducts an internal gap investigation to identify scientific measurement opportunities where an additional national payload could maximise the mission's scientific return. The objective is to determine where a complementary contribution would provide the greatest added value, not to identify deficiencies in the baseline mission.

Recommendation. A Gamma-Ray Spectrometer carried on the RAMSES main orbiter.¹

Rationale. Asteroid surfaces are altered by prolonged exposure to the space environment, a process that changes how the surface reflects light without changing what it is made of. Composition inferred from reflected light is therefore systematically biased, and the mission's payload measures composition entirely by reflected light.

Correcting that bias requires an independent measurement: one that is unaffected by the alteration and can be compared against the biased result. The confirmed payload does not carry one. Its correction would be derived from the same reflectance data it is meant to correct, which cannot be verified from within.

An elemental measurement supplies the missing reference. The alteration rearranges iron within a surface layer only micrometers thick; it does not add or remove atoms, so bulk elemental abundance is unchanged by it. A gamma-ray spectrometer measures that abundance from tens of centimeters below the surface, by a route that never passes through reflected light. Section 4 develops this in full.

Method. I constructed twenty measurement requirements from ESA's published objectives and checked each against all twelve confirmed instruments. Following my supervisors direction, I converged the analysis on Objective 2. Then, I registered four SSA opportunities as gaps and reduced them to one through three lenses and an overlap test, then decomposed the selected opportunity into a required observable before naming any instrument. Five candidates were evaluated against five criteria: three failed on architecture, and the remaining two separated based on the scientific complementarity.

Implementation: The instrument is obtained through a partnership with Johns Hopkins Applied Physics Laboratory and Lawrence Livermore National Laboratory, requesting a unit built to a design they have already flown.

The study draws on a vetted library of 52 sources, with NASA ADS bibcodes used wherever a confirmed abstract exists.

¹While close-proximity CubeSat deployment (e.g., Farinella) offers higher photon flux, the main orbiter provides the required SWaP (Size, Weight and Power) capacity for a higher-sensitivity, well-shielded detector to compensate for distance without payload reallocation.

Acknowledgement

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Above all, my deepest gratitude goes to my mom. Her continuous support and encouragement have been the true foundation of this journey. Without her, this achievement would not have been possible. Any remaining errors or omissions are entirely my own.

Contents

1	Introduction and Scope	6
1.1	Purpose	6
1.2	Scope Boundaries	6
2	Mission Overview	6
2.1	Mission Identity	6
2.2	Mission Objectives	6
2.3	Target and Timeline	7
2.4	Payload Architecture	7
3	Scientific Background: Space Weathering	8
3.1	The Process	8
3.2	Optical Consequences	8
3.3	Consequence for Compositional Retrieval	8
3.4	Why an Elemental Measurement Is Immune	9
3.5	Why This Matters Specifically at Apophis	9
4	Methodology	9
4.1	Two Governing Rules	9
4.2	Requirement Construction	9
4.3	Judgement	10
4.4	Opportunities and the Register	10
5	Instrument-to-Objective Coverage Matrix	10
5.1	Convergence on Objective 2	10
6	The Gap Register	11
7	Selection of the Priority Opportunity	11
7.1	Why the Gap Is Real	11
8	The Required Observable	12
9	Candidate Instruments and Trade Study	12
9.1	Evaluation Framework	12
9.2	Candidates	12
9.2.1	Gamma-Ray Spectrometer	12
9.2.2	Neutron Spectrometer	13
9.2.3	Mössbauer Spectrometer	13
9.2.4	MicrOmega Microscope	13
9.2.5	LIBS, ChemCam Class	13
9.3	Trade Study	13
10	Recommendation	14
10.1	Selection	14
10.2	Platform	14
10.3	Scope of the Measurement	14
11	Coupled Geochemical and Mineralogical Interpretation	14
12	Implementation Route	15
12.1	Heritage	15

12.2 Partnership	15
12.3 Items for the First Engagement	15
13 Sources and Citation Standard	16
13.1 Citation Standard	16
13.2 RAMSES Payload	16
13.3 Apophis and Space Weathering	16
13.4 Candidate Instruments	17
13.5 Gamma-Ray Spectrometry: Reference Design and Heritage	17
13.6 Official Agency Pages	18

1. Introduction and Scope

1.1 Purpose

This report is the complete deliverable of the first task of my summer training programme: an independent gap analysis of RAMSES (Rapid Apophis Mission for Space Safety), the joint ESA–JAXA planetary defence mission to near-Earth asteroid (99942) Apophis, carried through to a single instrument recommendation and a named route to obtaining it.

1.2 Scope Boundaries

Four boundaries apply.

- I work exclusively from the confirmed payload as finalised at ESA’s November 2025 Ministerial Council. Proposed instruments are out of scope.
- I declare no gaps (opportunities) from the coverage matrix alone. All are identified only through requirement derivation and capability checking against published instrument specifications.
- Any contribution must fit the established operational architecture: no new spacecraft, no new stage, no additional surface asset.
- The methodology is self-contained. Heritage missions are retained as context, not as the basis of gap judgements.

2. Mission Overview

2.1 Mission Identity

RAMSES is a joint ESA–JAXA planetary defence mission executed under ESA leadership with OHB Italia as prime contractor. The mission was formally selected and funded at ESA’s Ministerial Council in Bremen in late November 2025, bringing total contract value to approximately EUR 150 million. The spacecraft design derives from ESA’s Hera mission, adapted for a direct transfer trajectory to Apophis.

2.2 Mission Objectives

Four core objectives, verified against ESA’s published wording, structure both the mission and this analysis.

- **O1. Shape, orbit and rotation:** a before-and-after survey of Apophis’s shape, surface, orbit, rotation and orientation.
- **O2. Composition and interior structure:** determine asteroid composition, interior structure, cohesion, mass, density and porosity.
- **O3. Planetary defence and tidal response:** assess how the asteroid responds to Earth’s tidal forces.
- **O4. Solar System science:** scientific insight into the formation and evolution of the Solar System.

Parameter	Value
Designation	(99942) Apophis
Size	~375 metres across
Taxonomy	Sq type, LL ordinary chondrite analogue
Launch window	Spring 2028, H3 rocket, Tanegashima
Arrival at Apophis	February 2029
Close Earth flyby	13 April 2029, ~32,000 km

2.3 Target and Timeline

The flyby distance is about one-tenth of the way to the Moon and closer than satellites in geosynchronous orbit. While there is no risk of impact, the encounter is a rare scientific opportunity: a natural tidal experiment on an asteroid, observable at close range, not expected to repeat for a comparable object in the foreseeable future.

2.4 Payload Architecture

RAMSES carries a main orbiter payload plus two deployable 6U-XL CubeSats: Farinella, an orbiter CubeSat, and Don Quijote, a lander CubeSat. The twelve confirmed instruments across the three platforms are shown in the table below:

Instrument	Platform	What it measures
AFC ($\times 2$)	Main orbiter	Wide-field monochrome imaging for navigation and global shape reconstruction.
CHANCES	Main orbiter	High-resolution colour imaging; surface morphology and change detection across the encounter.
HAMLET	Main orbiter	Hyperspectral imaging, 650–1500 nm; mineralogical composition and space-weathering signatures.
TIRI	Main orbiter	Thermal infrared imaging; surface temperature, thermal inertia and regolith grain size.
RSE	Main orbiter	Radio tracking and inter-satellite links; mass, gravity field, spin state and tidal deformation.
RPS	Main orbiter	Energy and distribution of ions and electrons in the charged-particle environment.
HORUS	Farinella	Close-range imaging from low altitude; surface detail beyond the main orbiter's resolution.
Low-frequency radar	Farinella	Dielectric sounding of the interior; internal structure and heterogeneity at metre scale.
VISTA	Farinella	Thermogravimetric analysis of captured dust; volatile and organic content of released material.
SIA	Don Quijote	Surface seismometry; response to tidal stress, impacts and thermal cracking.
Gravimeter	Don Quijote	Local gravitational acceleration; density and internal mass distribution.
MARIE	Don Quijote	Three-axis surface magnetometry; response to Earth's magnetosphere during the flyby.

Table 1: The twelve confirmed instruments and the quantity each contributes.

3. Scientific Background: Space Weathering and Compositional Interpretation

This section establishes the physical problem I set out to address. The methodology that follows is built directly on it.

3.1 The Process

Airless bodies are continuously exposed to the space environment. Two processes dominate the alteration of their surfaces: bombardment by micrometeoroids, which melts and vapourises small volumes of regolith, and irradiation by the solar wind, which sputters and implants ions in the outermost grain surfaces.

Both processes reduce iron that is chemically bound within silicate minerals to metallic iron, which condenses as particles a few to a few tens of nanometres across. These particles accumulate in the rims of regolith grains and in the impact-produced glass that welds grains together. The population is referred to as nanophase metallic iron. Its abundance grows with the length of time a surface has been exposed, and the accumulated effect is described as the maturity of the surface.

The process is confined to a very thin layer. The altered rims are of the order of tens to hundreds of nanometres thick on individual grains, and the layer that reflected light samples is at most a few tens of micrometres deep.

3.2 Optical Consequences

Nanophase metallic iron is optically active out of proportion to its abundance. Three effects follow, and all three are well established observationally and in laboratory irradiation experiments.

- **Darkening.** Overall reflectance falls.
- **Reddening.** The continuum slope rises towards longer wavelengths.
- **Band suppression.** The depths of the diagnostic mafic absorption features near 1 and $2\mu\text{m}$ are reduced.

The combined result is that a weathered surface of a given composition presents a spectrum progressively less like that of the fresh material of the same composition. This is the origin of the long-standing discrepancy between the spectra of S-type asteroids and the ordinary chondrite meteorites they are believed to represent.

3.3 Consequence for Compositional Retrieval

Mineralogy is retrieved from reflectance spectra through a small set of parameters: the positions of the absorption band centres, the depths of those bands, and the ratio of their areas. Band centre positions are diagnostic of mineral chemistry; band depths and the band area ratio bear on the relative abundances of olivine and pyroxene.

Space weathering acts unevenly across these parameters. Band centre positions are only minimally displaced. Band depths and continuum slope, by contrast, are strongly affected, and any retrieval that depends on matching the overall shape of the spectrum is affected severely. A composition retrieved from a weathered spectrum without correction is therefore systematically, not randomly, in error.

Correcting the retrieval requires knowing how weathered the surface is. That quantity — the maturity — must itself be measured.

3.4 Why an Elemental Measurement Is Immune

The alteration redistributes iron; it does not create or destroy it. Iron that was bound in silicate lattices becomes metallic iron within the same volume of material. The total number of iron atoms per unit mass of regolith is unchanged, as is the abundance of every other element.

Bulk elemental abundance is therefore invariant under the process that biases the spectrum. Two conditions must hold for a measurement of it to serve as a reference. It must not itself be derived from reflected light, or it inherits the same bias. And it must sample material from below the altered layer, which is micrometres thick.

Gamma-ray spectroscopy satisfies both. Gamma rays emitted by naturally radioactive elements and by nuclei excited through cosmic-ray bombardment escape from tens of centimetres of material, four to five orders of magnitude deeper than the optical skin, and their detection involves no reflected light at any stage.

3.5 Why This Matters Specifically at Apophis

Apophis is classified as an Sq-type asteroid: spectrally intermediate between the S types, whose surfaces are mature, and the Q types, whose surfaces resemble fresh ordinary chondrite material. An Sq classification implies a surface that is partly, but not wholly, refreshed.

The mission's central scientific premise is that the close terrestrial encounter of 13 April 2029 will disturb the surface. If it does so sufficiently, it will expose unweathered material and reset the maturity clock. Whether an encounter of this severity is sufficient remains an open question in the literature.

The weathering state of Apophis is therefore not a nuisance parameter to be removed. It is a measurement target in its own right, and one of the few for which the 2029 encounter provides a natural experiment that will not recur.

4. Methodology

4.1 Two Governing Rules

I set two rules before beginning, and they account for most of what follows.

A gap is what the mission never set out to do, not where it fell short. A requirement is derived from ESA's own objectives, so the confirmed payload was designed against it. What remains uncovered there is a margin of precision on a demand the mission already answers, which opens a partial opportunity at most, never a gap. Registering it as a gap would assert that the mission failed a demand it set for itself, and I found no evidence supporting that claim.

Matching ESA's data is validation; changing it is correction. Only the second constitutes a contribution. An instrument that reproduces a measurement the mission already makes adds a dataset; one that alters how an existing measurement must be read adds a condition on the mission's own results. I admitted only the latter.

4.2 Requirement Construction

Each objective is decomposed into testable measurement requirements traceable to ESA's published wording. A requirement is built from four terms that make it precise enough to fail: one narrow scientific purpose, one physical observable that serves it, one precision level where precision is decisive, and one boundary stating what the requirement excludes. A bare statement such as *measure the shape* is satisfied by any camera and exposes nothing.

I constructed twenty requirements across O1–O3. O4 produces no original requirements: formation and evolution is not directly measurable and inherits interpretively from O1 and O2.

4.3 Judgement

The check is binary and repeatable. A requirement is *met* if at least one confirmed instrument measures the required quantity itself, at the scope demanded, with a published source stating the capability; *not met* if none does. Every requirement is checked against all twelve instruments without exception. The coverage matrix guides attention but never filters the check. A proxy or adjacent measurement does not close a requirement.

4.4 Opportunities and the Register

Beyond the requirements derived from ESA’s objectives, I identify SSA opportunities: measurement openings the mission never set out to address. My supervisor directed that these be named opportunities rather than requirements, since calling them requirements would imply that ESA’s objectives demand them. I register them as gaps only where no confirmed instrument meets them.

Each registered gap carries an eight-field card: gap identifier, linked requirement or opportunity, parent objective, evidence, scientific impact, SSA strategic fit, supporting sources, and status.

5. Instrument-to-Objective Coverage Matrix

Rows are the twelve confirmed instruments; columns are the four mission objectives. Each cell answers one question: does this instrument contribute at least one measurement that serves this objective? The verdicts are my own reasoned judgements, each backed by a published source, and not an official ESA or JAXA assessment.

The matrix is a hypothesis generator, not a gap detector. A False cell is a lead — a place to look — never a gap in itself, because absence at instrument level does not entail absence at measurement level.

Instrument	O1 Shape / Orbit	O2 Composition	O3 Tidal / Defence	O4 Solar System
AFC (×2)	T	F	T	T
CHANCES	T	T	T	T
HAMLET	F	T	T	T
TIRI	F	T	T	T
RSE	T	T	T	T
RPS	F	F	T	F
HORUS (Farinella)	T	F	F	T
Radar (Farinella)	F	T	T	T
VISTA (Farinella)	F	T	T	T
SIA (Don Quijote)	F	T	T	T
Gravimeter (Don Quijote)	F	T	T	T
MARIE (Don Quijote)	F	F	T	T
Column total (True)	4	8	11	11

Totals: 34 True and 14 False across 48 cells.

5.1 Convergence on Objective 2

Checking all twelve instruments against all twenty requirements at uniform depth reproduces ESA’s own trade study; an exploratory pass returned almost entirely *met*. On supervisor direction, I converged the analysis on Objective 2 — composition and interior — which carries the resource-assessment lens relevant to the Agency’s stated interest. ESA’s four objectives are unchanged; only the depth of my analysis is converged.

6. The Gap Register

Four opportunities entered the register.

ID	Opportunity	Why no confirmed instrument meets it
GAP-1	Hydration state	HAMLET's range ends at 1500 nm. The diagnostic OH and water bands near 2.0 and 3.0 μm fall outside it, and no other instrument covers that region.
GAP-2	Weathering-corrected composition	All compositional instruments observe only the space-weathered surface layer. No independent reference exists against which the weathering correction can be established.
GAP-3	Regolith grain-size distribution	Only TIRI bears on grain size, through thermal inertia, which is a proxy. Under the study's proxy rule a proxy does not close a requirement.
GAP-4	Free Fe-Ni metallic phase fraction	MARIE and HAMLET each give indirect indications of metal content, but neither separates free metallic iron from iron bound in silicates or oxides.

7. Selection of the Priority Opportunity

I passed the four registered opportunities through three lenses in fixed order, and added a fourth test because the engineering lens did not separate the survivors.

Lens	Question and verdict	Outcome
Strategist	Is it unique to the 2029 encounter, and does it fit the architecture? Free-metal content is fixed in time and measurable at any body, so a contribution there is substitutable. Hydration, read as a surface property, requires a 2–3 μm spectrometer on a platform the mission does not carry.	GAP-4, GAP-1 out
Scientist	Is the answer already known? An LL-analogue body is expected dry and metal-poor, so both would most likely return upper limits. Two lenses reaching the same pair on unrelated grounds is a result in its own right.	Same two, confirmed
Engineer	Does it fit the platform? Neither survivor eliminated.	No cut
Overlap test	Is the question already in the baseline? Regolith movement under tidal stress is partly addressed by R3.2, which compares surface imagery before and after the encounter, so GAP-3 sharpens the scale of an existing answer. Weathering-corrected composition is touched by no requirement and no confirmed instrument.	GAP-3 out

GAP-2, weathering-corrected composition, is selected as an Opportunity.

7.1 Why the Gap Is Real

As set out in Section 4, correcting a weathered spectrum requires knowing the maturity of the surface. HAMLET is expected to yield a maturity map, which invites the objection that the quantity is already measured. The derivation is closed: verifiable only against the model that produced it, with no external reference.

Overlap at the surface is not disqualifying. It is the biased term of the difference, and without it there would be nothing to correct. Of the confirmed payload, only HAMLET makes a claim on the same ground, and that claim is closed.

Instrument	Verdict	Basis
HAMLET	Closed inference	Maturity map derived from the spectrum it would correct.
TIRI	Useful	Supplies grain size, so this study need not measure it.
Low-frequency radar	Unrelated	Olivine and pyroxene are both insulating; the dielectric measurement is blind to nanophase iron by orders of magnitude, at ~ 20 m vertical resolution.
CHANCES	No overlap	Locates where fresh material may be; never quantifies it.
AFC and PALT	No overlap	Geometry, treated as a design constant.

8. The Required Observable

The gap sets what must be measured; only then did I begin the search for a candidate.

Required observable: quantitative elemental abundance of the near-surface regolith. Mass fractions and ratios measured over the body from orbit, by a route that does not pass through reflectance.

- **Why this quantity.** As established in Section 4.4, elemental abundance is invariant under the process that biases the spectrum, and can therefore stand outside the inference as a reference.
- **Why from orbit.** The correction is applied to maps. A reference taken at a single landing site cannot travel across a body; one measured on the same coverage as the spectral data can.
- **Why below the rind.** The sampling depth must exceed the altered layer, or the reference carries the same bias it is meant to correct. This is not a preference for depth as such: below the layer that reflected light samples, the alteration biases nothing.

9. Candidate Instruments and Trade Study

9.1 Evaluation Framework

Each candidate is read in two sections. **Scientific relevance** asks which observable the instrument provides and how directly it addresses the identified gap. **Mission suitability** asks five questions: architecture compatibility, scientific complementarity, Technology Readiness Level, operational complexity and power consumption.

I treat architecture as a screening criterion, not a differentiator. A candidate is scored only until the criterion that removes it: an instrument that cannot be operated on this mission is not compared further, and I record its remaining fields as not evaluated rather than scoring them.

9.2 Candidates

9.2.1 Gamma-Ray Spectrometer

Scientific relevance. Provides elemental abundance of the near-surface regolith from natural and cosmic-ray-induced gamma emission, at a sampling depth of tens of centimetres. This is the required observable, measured by a route independent of reflectance and from a depth beyond the altered layer.

Mission suitability. Fits the orbiter with no change to the mission concept and no surface asset. Complements the mineralogical dataset: elemental closure narrows the set of mineral models a spectrum admits. High Technology Readiness Level, flown at the Moon, Mars, Mercury, Eros, Vesta and Ceres, and in flight to Psyche. Low operational complexity, passive and continuous, without requiring a robotic arm or any other mechanism. Modest power, with a cryocooler required for the germanium sensor.

9.2.2 Neutron Spectrometer

Scientific relevance. Provides bulk hydrogen content and mean atomic mass of the regolith to a comparable depth. Indirect for this gap: hydrogen does not enter the mineralogical inversion, so the result cannot narrow the model set.

Mission suitability. Fits the orbiter on the same terms as the Gamma-Ray Spectrometer, with comparable maturity, complexity and power. It separates on scientific complementarity: neutron spectroscopy primarily constrained hydrogen and volatile abundance, whereas resolving this gap requires multi-element ratios (such as Mg, Si, and Fe) to constrain mineralogical models.

9.2.3 Mössbauer Spectrometer

Scientific relevance. Provides iron oxidation state and phase distribution, including nanophase metallic iron, to about $200\ \mu\text{m}$. Its two emission lines at 6.4 and 14.4 keV give intrinsic depth selectivity requiring no mechanism.

Mission suitability. Contact instrument. On supervisor direction I established that sub-surface observation does not require a landing, which removed the premise under every contact candidate: the contribution is orbiter-only, so this instrument cannot be operated. Removed at architecture.

9.2.4 MicrOmega Microscope

Scientific relevance. Provides grain-scale mineralogy and texture across $0.5\text{--}3.65\ \mu\text{m}$ at $20\ \mu\text{m}$ sampling. Direct if unaltered grains are exposed: it supplies the fresh spectrum rather than a correction to the weathered one.

Mission suitability. Contact instrument, with access dependent on surface disturbance. Removed at architecture.

9.2.5 LIBS, ChemCam Class

Scientific relevance. Provides elemental composition as a function of depth, the laser removing the rind as it measures. It solves access better than any other candidate and still returns the wrong quantity. By the argument of Section 4.4, the alteration barely moves bulk elemental abundance, so profiling elements through the rind would most likely return no difference.

Mission suitability. Close-range instrument requiring high pointing precision. Removed at architecture.

9.3 Trade Study

Candidate	Arch.	Compl.	TRL	Cplx.	Power
Gamma-Ray Spectrometer	Fits	Complements	High	Low	Modest
Neutron Spectrometer	Fits	Partial	High	Low	Modest
Mössbauer	Requires lander	—	High	—	—
MicrOmega microscope	Requires lander	—	High	—	—
LIBS, ChemCam class	Requires lander	—	High	—	—

10. Recommendation

10.1 Selection

Selecting a Gamma-Ray Spectrometer as the preferred payload enhancement, carried on the RAMSES main orbiter. It reads elements rather than light, from tens of centimetres down, below the layer the weathering alters.

My case does not rest on a deficiency in the RAMSES baseline. The mission's weathering correction is drawn from the measurement it corrects, with nothing outside to check it against. This supplies that reference, conditioning how the existing hyperspectral dataset is read rather than adding a dataset beside it, and does so within the established operational architecture: no new spacecraft, no new stage, no surface asset.

10.2 Platform

The main orbiter, not Farinella. The cooled germanium sensor exceeds what a 6U-XL CubeSat carries in mass and standing power, and the measurement requires an accumulation window measured in months. It also belongs beside HAMLET: sharing a platform means sharing observation geometry and coverage, which is what permits one map to be subtracted from the other.

10.3 Scope of the Measurement

Two properties bound the measurement, and both describe the instrument's role in the payload rather than detract from it.

Spatial resolution. The instrument reads a broad patch, tens to hundreds of metres. The mission already maps in detail; this supplies the calibrated value that map is read against.

Integration time. Gamma emission is faint, so precision accumulates over hours and days of counts. The instrument is passive, with no laser, mechanism or pointing, and gathers while the spacecraft works.

11. Coupled Geochemical and Mineralogical Interpretation

The Gamma-Ray Spectrometer provides an independent elemental constraint that complements HAMLET's mineralogical observations. The combined dataset reduces uncertainty in compositional interpretation while recognising that elemental abundances and mineralogy remain linked through inversion models.

The interpretation chain runs as follows. Spectral inversion of HAMLET data yields a set of candidate mineralogical models, several of which fit one spectrum. Elemental closure rejects those inconsistent with the measured abundances. The admissible set narrows.

Because HAMLET's spectral range excludes the $2\mu\text{m}$ pyroxene absorption band, spectral inversions suffer from severe compositional degeneracy. Independent elemental abundances (e.g., Fe/Si and Mg/Si ratios) break these degeneracies, providing the necessary mathematical boundary conditions to isolate unique mineralogical fits.

12. Implementation Route

12.1 Heritage

Planetary gamma-ray spectroscopy is not a technology to be developed. It has flown to the Moon, Mars, Mercury, Eros, Vesta and Ceres, and is in flight to Psyche. The question is not whether the instrument can be built but through whom.

12.2 Partnership

I close the study by naming how the instrument reaches the mission: a partnership with Johns Hopkins Applied Physics Laboratory and Lawrence Livermore National Laboratory, requesting a unit built to a design they have already flown — a flight spare, an engineering model, or a repeat of the Psyche design.

Between them, these two laboratories have built every high-purity germanium gamma-ray spectrometer flown on a planetary mission.

Laboratory	Role	Gamma-ray spectrometers built
Johns Hopkins APL	Instrument lead and integrator	NEAR Shoemaker, MESSENGER, MEGANE, Psyche, Dragonfly
Lawrence Livermore NL	High-purity germanium sensor	MESSENGER, MEGANE, Psyche

MEGANE is this proposal in structure: one agency contributing a gamma-ray spectrometer to another agency's small-body mission, built by an experienced laboratory and selected through a mission-of-opportunity call. Its cryocooler is a catalogue-derived unit qualified by the instrument builder, as on MESSENGER, so cooling is not a development item.

12.3 Items for the First Engagement

Two engineering items remain open and belong in the first technical exchange: the footprint achievable at the planned operating altitude, and boom against body mounting, since spacecraft materials generate the same background the instrument measures. Both organisations named are United States national laboratories, so export-control and technology-transfer conditions apply and belong in the first conversation rather than the last.

With the gap identified, the observable defined, the instrument selected and the route to obtaining it named, the study is complete.

13. Sources and Citation Standard

13.1 Citation Standard

The citation standard is the NASA ADS bibcode: I reference every per-instrument claim to an ADS-indexed abstract wherever a confirmed one exists. Where no dedicated ADS paper could be confirmed at the research date, the publisher or the official instrument or agency page is cited instead, with an explicit note rather than a substitute attribution.

13.2 RAMSES Payload

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13.3 Apophis and Space Weathering

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